

The Climate Cost of Computing

Quantifying Climate Value at Risk for Global Data Center Infrastructure

Rémi Paccou
Thomas Epelbaum

March 2026

Foreword

Steve Witwhite

EVP, Secure Power Division, Schneider Electric



The data center industry stands at an inflection point. Over the past three decades, we have built the physical layer of the digital economy—facility by facility, market by market—into an infrastructure class that now underpins every sector of the global economy. The scale of what comes next dwarfs what came before: the AI era demands not incremental expansion but a structural transformation of how we design, power, and operate these critical assets.

At Schneider Electric, we work alongside operators, investors, and grid planners who are making capital allocation decisions today that will determine the resilience of digital infrastructure through 2050. The question they face is not whether climate risk affects data centers—every operator who has managed through a winter storm, a heatwave, or a grid emergency knows the answer. The question is how much, where, and what to do about it.

This paper provides the first quantitative answer. By translating physical and transition climate hazards into enterprise value loss at facility level—across the global installed base and the prospective AI pipeline—it makes visible a dimension of risk that capital markets have not yet priced. The supply chain analysis is particularly consequential: the finding that indirect business interruption exceeds direct exposure by an order of magnitude reframes how we think about facility resilience.

What gives this work practical weight is that it does not stop at diagnosis. The adaptation analysis demonstrates that structured investment reduces climate exposure by a third, with positive returns under every scenario tested. For an industry deploying trillions in long-lived assets, that is an actionable result.

The analytical framework developed here—and the Schneider Electric Sustainability Research Institute's commitment to open, reproducible methodology—provides the foundation for integrating climate risk into the investment decisions that will shape our industry for the next quarter century.

Executive Summary

Rémi Paccou

Schneider Electric Sustainability Research Institute | Doctoral Researcher, CIRED

Data centers are the physical infrastructure of the digital economy—and the only major asset class whose climate exposure has never been financially quantified. The sector requires US\$3–7 trillion in new investment by 2030. These assets carry 15–25 year operational horizons, yet are being built without pricing the climate trajectory they will operate under.

The protection gap quantifies the consequence: of US\$296 billion in global natural catastrophe losses in 2025, US\$167 billion went uninsured—a gap growing at 6–10% annually. For data centers, placement programs already exceed US\$1 billion; premiums are projected to triple by mid-century.

Climate risk for this asset class is *recognized but not priced*. Three structural gaps explain why: no framework translates hazards into enterprise value loss; no analysis covers both the installed base and the AI-era pipeline; no methodology decomposes risk into actionable channels.

This paper closes all three gaps. It introduces Climate Value at Risk (ClimVaR)—a facility-level valuation framework integrating five risk channels through discounted cash flow analysis, Shapley decomposition for hazard attribution, and Leontief input-output analysis for supply chain propagation. Applied to 8,572 installed data centers across 120 countries and three AI growth scenarios, it produces the first global financial quantification of climate risk for digital infrastructure.

US\$388B

ClimVaR on installed base

38%

Average intensity (of enterprise value)

×11

Supply chain to direct BI ratio

30–39%

Adaptation value across all scenarios

Four Findings

1. Aggregate Exposure

The installed base carries US\$388B in climate-adjusted value erosion (38% of enterprise value). With AI infrastructure, exposure rises to US\$1.0–3.7T.

2. Geographic Dispersion

Risk intensity ranges from 28% (US) to 56% (China). The dominant channel shifts by region: BI in the US, carbon in Europe, damage in APMEA.

3. Structural Inversion

Legacy assets face a carbon problem (72% transition risk). AI-era assets face a physics problem (90%+ physical). Each GW of AI capacity carries $\times 2.6$ the physical risk of installed base.

4. Adaptation Value

Proactive adaptation protects US\$150B–1.1T depending on scenario. Selective (one action per facility) captures 56% of Proactive value at lower cost.

What This Means

For investors: a portfolio with 38% ClimVaR intensity requires repricing. **For operators:** the Shapley decomposition identifies which channel to address first at each facility. **For insurers:** the $\times 11$ supply chain multiplier explains why facility-level loss ratios

underestimate systemic exposure. **For policymakers:** the concentration of AI investment in climate-vulnerable locations represents a market failure requiring climate-adjusted siting rules.

Executive Summary	2	Chapter III	
		Results	12
		Aggregate Exposure	12
		Geographic Dispersion	12
		AI Amplification	13
		Adaptation Value	13
		Case Studies	14
Chapter I		Chapter IV	
Introduction	6	Discussion	15
Context and Urgency	6		
The Unpriced Gap	7	Chapter V	
Literature and Positioning	7	Conclusions & Future Research	16
Contributions	8		
		References	17
Chapter II		Appendices	18
Methods & Data	9	A. Supply Chain Propagation	19
The Five Risk Channels	9	B. Risk Channel Decomposition	20
Shapley Decomposition	10	C. Adaptation as Portfolio Optimization	21
Supply Chain Propagation	10	D. Data & Scenario Hypotheses	22
Scenarios and Pathways	11		
Data	11		

Chapter I

Introduction

Context and Urgency

Digital Infrastructure Under Climate Stress

The winter of 2025–2026 illustrated the financial consequences of grid stress under extreme weather at data center scale. In January 2026, a winter storm paralyzed the eastern United States. In Virginia’s “data center alley”—home to the world’s largest concentration of digital infrastructure—wholesale electricity prices surged from \$200 to \$1,800 per MWh overnight, a ninefold increase. The PJM Interconnection, serving 67 million people across the US Mid-Atlantic, approached 147 GW of demand, shattering the previous winter record of 143.7 GW set one year earlier. The Department of Energy issued emergency orders requiring data centers to prepare backup generation.

This was not an isolated shock. In the same winter, Storm Éowyn battered Northern Europe with gusts exceeding 180 km/h, knocking out power to over one million customers across Ireland and the United Kingdom and disrupting connectivity in data center clusters serving pan-European cloud workloads (Gallagher Re, 2026). In China, Typhoon Yagi—the most powerful western Pacific storm of 2024—had months earlier flooded Guangdong province, home to one of Asia’s densest concentrations of digital infrastructure, causing cascading grid failures that took 72 hours to resolve. In the Middle East, record temperatures exceeding 50°C pushed cooling systems to their

design limits at facilities in Dubai and Riyadh.

What these episodes demonstrate is that physical climate hazards now impose direct financial costs on the infrastructure that supports the digital economy. Data centers are capital-intensive, long-lived assets whose financial performance depends on energy systems, physical site conditions, and insurance markets—all of which climate change is repricing simultaneously. Four phenomena define this operating environment.

Compute as strategic infrastructure

Data centers have transitioned from commercial real estate to strategic national infrastructure. In June 2025, the United States formally designated hyperscale data centers as Critical Digital Infrastructure, enabling expedited permitting and eligibility for federal risk-sharing programs. By 2027, an estimated \$8.75 trillion of global economic activity will depend directly on data center operations (Equinix, 2025). Geopolitical competition reinforces this shift: at the February 2025 AI Action Summit in Paris, the EU announced InvestAI, a €200 billion initiative for AI infrastructure. Private investment in US data centers reached record highs in 2024, with hyperscalers Amazon, Microsoft, and Google committing approximately \$210 billion in AI-related cap-

ital expenditure. Morgan Stanley projects nearly \$3 trillion in global data center spending by 2029; McKinsey estimates cumulative outlays approaching \$7 trillion by 2030.

China operates at comparable scale through state-backed programs, with over 25 GW of installed capacity concentrated in the Yangtze Delta, Pearl River Delta, and Beijing-Tianjin corridor. China's "East Data West Compute" initiative redistributes capacity toward western provinces—a strategy that reduces latency costs but increases exposure to seismic and water-stress hazards. The Gulf states deploy sovereign AI infrastructure in regions where ambient temperatures already exceed cooling design thresholds for much of the year. Across all markets, these assets will operate through mid-century under climate conditions that current siting and valuation frameworks do not price.

A decade of climate events disrupting operations

The vulnerability of digital infrastructure to physical climate hazards is documented and escalating. Hurricane Sandy (2012) flooded Manhattan basement generators—Datagram reported five feet of standing water; Cogeco Peer 1 mobilized bucket brigades to carry diesel by hand. Hurricane Harvey (2017) tested Houston facilities at design limits with over 40 inches of rainfall; survival hinged on pre-emptive fuel sourcing and elevated siting, not architectural margin. Winter Storm Uri (2021) introduced a different failure mode: temperatures 30–40°F below seasonal norms caused generator fuel to gel, batteries to fail, and cooling systems to malfunction. NERC concluded that the ERCOT grid came within minutes of uncontrolled cascade failure.

The July 2022 UK heatwave pushed ambient temperatures above 40°C for the first time in recorded history. Google's London data center and Oracle's UK facility reported cooling-related outages—not because equipment failed, but because ambient conditions exceeded the thermal envelope within which mechanical cooling was designed to operate. Global data center electricity consumption reached approximately 450 TWh in 2024, growing at nearly 20% annually (IEA, 2025). At this scale, the interaction between demand growth and physical climate hazards is a permanent feature of the sector's operating environment.

Insurance retreat and the protection gap

Physical hazards damage assets. The investment problem arises from how risk is priced, absorbed, or refused by insurance markets. The global insurance protection gap for natural

catastrophes stood at 56% in 2025: of \$296 billion in economic losses, \$167 billion went uninsured (Gallagher Re, 2026). This figure is not a short-term fluctuation. Global uninsured losses have grown at 6–10% annually in real terms between 2015 and 2024 (Swiss Re, 2025). In the EU, only 20% of natural catastrophe losses are covered; in China, penetration remains below 15%.

When insurance retreats, asset values follow. Evidence from the US residential market documents declines of 12–48% in property values following insurance non-renewal (WWF, 2025). More than 1.9 million home insurance policies have been non-renewed in the United States since 2018. The WEF, drawing on S&P data, projects climate-driven costs for data centers at \$81 billion annually by 2035, rising to \$168 billion by 2065, with cumulative losses of \$3.3 trillion by 2055 under high-emissions scenarios. XDI warns that global insurance premiums for data centers could triple or quadruple by mid-century. For data center construction, Allianz Commercial notes that placement programs already routinely exceed \$1 billion, with mega-campuses seeking up to \$3 billion in coverage syndicated across multiple insurers.

When private insurance retreats, governments absorb part of the residual risk. The United States spent approximately \$1 trillion (3% of GDP) on disaster recovery and climate-related needs in the twelve months to May 2025. France's public catastrophe scheme has raised premium surcharges from 12% to 20% starting in 2025, with reinsurer CCR anticipating a 40% rise in claims by 2050.

The design window is closing

Between 40% and 60% of 2035 global data center capacity will be built in the next five years. Once commissioned, these facilities cannot relocate. Every siting decision made today embeds a 25-year bet on climate trajectories. The available evidence points to substantial returns on adaptation investment: every dollar invested in climate resilience can avoid up to \$13 in losses in the United States; each £1 spent on flood risk management avoids £8 in damages in the United Kingdom (WWF, 2025). Yet current frameworks for data center siting continue to optimize for power availability, fiber connectivity, latency, and near-term costs. The global project pipeline concentrates investment in markets selected for land availability and energy access rather than climate resilience. Physical bottlenecks already identified—grid connection timelines, transformer supply chains with 18–36 month lead times, cooling water availability—are precisely the vectors through which climate hazards translate into asset-level financial risk.

The Research Gap

These converging pressures raise a question: **how much climate risk does the global data center sector actually carry?**

Academic literature

Despite the sector's strategic importance and capital intensity, academic research quantifying climate risk for data centers is sparse. Our systematic review (detailed in Appendix D) identified only one sustained research program directly addressing data center climate exposure—the work of Pasupuleti and co-authors—compared to hundreds of studies for analogous infrastructure sectors. Pasupuleti (2022) integrates hydrologic simulation with cloud workload models to quantify impacts on service availability and financial loss. Pasupuleti (2024) extends this into a climate-aware GIS methodology for site selection, finding that hazard-aware siting reduces expected annual damage by 63%. Pasupuleti (2025) demonstrates that the cost premium for resilient locations can be offset by reduced disruption over asset lifetimes. This program establishes foundations for flood-aware site selection but does not attempt portfolio-level assessment, does not cover the full range of risk channels (cooling, carbon, supply chain), and does not address the prospective AI-era pipeline.

A substantial body of work on analogous critical infrastructure provides transferable methodologies, though none has been applied to data centers. Forzieri et al. (2018) project that damages to European critical infrastructure could exceed ten times present levels (€3.4 billion/year) by end of century. Dawson et al. (2018) provide a UK national systems framework identifying that increased interdependence between energy and ICT systems amplifies risk cascades. Oughton et al. (2019) analyze 7.6 million mobile assets globally, estimating \$2.69 billion in coastal flood damage under RCP 8.5. No peer-reviewed study documents insurance protection gaps, premium trends, or market withdrawals specifically for data centers. Quantitative valuation frameworks exist for energy infrastructure (In et al., 2020–2022; Prime, 2018; Pasha, 2025) but none addresses data center-specific vulnerabilities: cooling system dependencies on ambient temperature, power quality requirements, and the asymmetric revenue impact of service interruptions relative

to physical damage.

Practitioner analyses

XDI's 2025 Global Data Centre Physical Climate Risk Report covers 8,868 facilities across eight climate hazards with engineering vulnerability functions calibrated to data center archetypes. Its Maximum Value at Risk metric quantifies structural damage as a fraction of replacement cost. By design, the report does not assess extreme heat—the dominant economic channel for facilities where cooling accounts for 30–50% of operating costs—and excludes transition risk and the AI-era pipeline.

The WEF, drawing on S&P Global data, produced the first sector-aggregate financial projection: \$3.3 trillion in cumulative climate costs by 2055. This estimate established financial materiality at board and regulatory level but operates at aggregate level without facility-level breakdowns or channel decomposition. Verdantix's December 2024 benchmarking of 19 physical climate risk vendors identified adaptation planning and supply chain risk quantification as the two weakest capabilities across the entire market—confirming that the analytical dimensions most relevant to capital allocation are precisely those least developed.

Three structural gaps

First, existing analyses measure physical hazard intensity or replacement cost, not enterprise value loss—the metric that governs capital allocation, TCFD disclosure, and insurance underwriting. Second, all existing analyses characterize the installed base; none models the prospective AI-era pipeline that represents over 80% of projected capacity additions—the assets whose climate exposure is determined by decisions being made now. Third, no framework decomposes aggregate risk into actionable economic channels: without knowing whether a given facility's ClimVaR is driven by cooling costs, carbon pricing, physical damage, or supply chain disruption, adaptation budgets cannot be allocated with analytical rigor. Verdantix documents this gap across all 19 vendors it benchmarks; none closes it.

Contributions

This paper addresses all three gaps. It asks: *What is the Climate Value at Risk for global data center infrastructure, how does it vary across locations and scenarios, and which adaptation strategies generate positive risk-adjusted returns?*

It makes four contributions. **First**, it adapts the ClimVaR methodology (Epelbaum, 2025) to data center infrastructure, embedding physical and transition climate risks into discounted cash flow analysis through five sector-specific channels (γ cool-

ing, $CC^{1,2,3}$ carbon, β interruption, δ damage, ξ productivity). Two mathematical tools—Shapley decomposition for hazard attribution and Leontief input-output analysis for supply chain propagation—bring capabilities not previously applied to this sector.

Second, it assembles a geocoded database of 8,572 installed facilities and up to 4,633 prospective AI-era facilities under three growth scenarios, linked to 29 documented financial parameters, downscaled CMIP climate projections, and country-level carbon price trajectories.

Third, it constructs a scenario architecture crossing warming pathways (RCP 2.6, 4.5), transition policies (BAU, Net Zero), AI growth trajectories (Limits to Growth, Sustainable AI, Abundance Without Boundaries), and three adaptation strategies (Baseline, Selective, Proactive)—identifying which interventions generate positive returns, under which conditions, and in which markets.

Fourth, it quantifies the business case for adaptation, demonstrating that proactive strategies reduce aggregate ClimVaR by 30–39% with positive net present value under every scenario tested.

Data centers host the computational infrastructure used for climate projections and risk pricing across other sectors, and store the data underpinning TCFD and CSRD disclosures. Yet the sector's own financial exposure to climate trajectories remains unquantified. The risk is not unrecognized—it is unpriced.

Paper Structure

Chapter II develops the ClimVaR framework with five risk channels, Shapley decomposition, Leontief supply chain propagation, and the scenario architecture. **Chapter III** presents four

findings at macroeconomic scale—aggregate exposure, geographic dispersion, AI amplification, adaptation value—and three facility case studies. **Chapter IV** interprets these results against existing literature and draws implications for operators, investors, insurers, and policymakers. **Chapter V** summarizes contributions, acknowledges limitations, and outlines extensions.

Chapter II

Methods & Data

The ClimVaR Framework

The framework rests on three analytical layers. First, a climate-adjusted DCF model translates physical and transition hazards into enterprise value loss through five sector-specific risk channels. Second, Shapley decomposition attributes business interruption risk to individual hazards. Third, Leontief input-output analysis traces how climate disruptions propagate through supply chains. The integration of these tools responds to a specific analytical need: standard climate risk assessments produce aggregate scores—useful for screening but insufficient for investment decisions, which require knowing *which* hazards drive exposure and *where* in the value chain risk originates. Shapley and Leontief provide precisely these capabilities. Both are grounded in foundational contributions to mathematical economics—Shapley (Nobel 2012) for cooperative game theory, Leontief (Nobel 1973) for input-output analysis—but neither has been applied to data center infrastructure.

Climate-adjusted DCF

Infrastructure valuation conventionally relies on discounted cash flow analysis, estimating present value as the sum of future cash flows discounted for time and risk. For a facility generating free cash flows $FCF^{(t)}$ over horizon T with weighted average cost of capital ω :

$$V = \sum_{t=1}^{T+1} \frac{FCF^{(t)}}{(1 + \omega)^t}$$

Climate change affects these cash flows through multiple channels. We embed these effects directly into valuation, holding the discount rate constant—making ClimVaR a conservative lower bound on total value erosion:

$$\text{ClimVaR} = V - V^{cc} = \sum_{t=1}^{T+1} \frac{FCF^{(t)} - FCF^{(t,cc)}}{(1 + \omega)^t}$$

The full valuation equation aggregates revenue losses, COGS increases, and OPEX increases across five channels:

$$\text{ClimVaR} = \sum_{t=1}^{T+1} \frac{(1 - \tau)}{(1 + \omega)^t} \left[op \cdot \Delta Y + Y^{cc} (s_o \xi^{(t)} + e_c \gamma^{(t)}) + \sum_{i=1}^3 CC^{(t,i)} + (1 + \iota) \cdot AV \cdot \delta^{(t)} \right]$$

where τ is the tax rate, op the operating margin, ΔY the revenue loss from business interruption, s_o the outdoor salary ratio, e_c

the cooling energy ratio, ι the insurance loading factor, and AV the asset value.

Exhibit 2. ClimVaR framework: climate drivers propagate through five risk channels to affect enterprise value. Cooling costs (γ) and carbon costs (CC) are particularly significant for data centers given their energy intensity.



Source: Adapted from Epelbaum (2025). Five risk channels map physical and transition drivers to enterprise value loss through OPEX, revenue, and CAPEX impacts.

The Five Risk Channels

Each channel captures a distinct mechanism. **Cooling stress (γ):** data centers consume 30–50% of facility energy for cooling. Each 1 °C temperature increase raises cooling costs 2–4%, making γ unusually prominent relative to other asset classes. **Carbon pricing (CC^{1,2,3}):** energy intensity 10–100× that of office buildings means Scope 2 (purchased electricity) typically represents 70–85% of total emissions. Trajectory-dependent carbon prices under BAU and Net Zero pathways drive this channel. **Business interruption (β):** extreme weather causes revenue loss proportional to downtime probability, amplified by supply chain propagation. 99.99% availability SLAs mean out-

ages translate directly into contractual penalties. **Physical damage (δ):** acute hazards (flood, wind, wildfire) destroy physical assets, reducing residual value and triggering insurance cost increases. **Heat productivity loss (ξ):** outdoor labor productivity declines affect construction and maintenance costs—generally small for automated facilities but material at sites with recurrent heat waves.

Two channels defined in the general framework—customer shift (σ) and nature capital (ρ)—are not parameterized in this analysis due to insufficient calibration data, but remain in the mathematical architecture for future activation.

Shapley Decomposition

Business interruption aggregates exposure to multiple climate hazards: floods, droughts, storms, wildfires, extreme heat. A facility facing both drought and heat risk experiences compound effects—the joint impact exceeds the sum of individual impacts. Simple additive decomposition produces attributions that either fail to sum to total risk or depend arbitrarily on ordering. The Shapley value resolves this by averaging marginal contributions across all possible orderings of hazard inclusion:

$$\phi_h(v) = \sum_{S \subseteq H \setminus \{h\}} \frac{|S|!(|H| - |S| - 1)!}{|H|!} [v(S \cup \{h\}) - v(S)]$$

The decomposition satisfies four axioms: efficiency (attributions sum to total risk), symmetry, null player (irrelevant hazards receive zero attribution), and additivity. With $|H| = 5$ hazards, exact computation requires $2^5 = 32$ coalition evaluations per facility—tractable for the full database.

Shapley decomposition applies *within* the β channel, where hazard interactions are significant. Across channels, ClimVaR components aggregate additively: cooling, carbon, damage, and productivity enter through distinct cash flow lines and do not interact mechanically.

Climate risk for data centers is not a single hazard—it is a portfolio of interacting exposures. The Shapley decomposition reveals which channels drive value erosion and where adaptation generates the highest marginal return.

Supply Chain Propagation

Facility-level analysis misses indirect exposure through upstream supply chains. A flood affecting semiconductor fabrication in Taiwan, a hurricane damaging transformer manufacturing on the Gulf Coast, or a drought constraining equipment production—all translate into business interruption risk for facilities thousands of kilometers distant.

We use Leontief input-output analysis to trace propagation. Let A denote the technical coefficient matrix from OECD Multi-Regional Input-Output Tables (2023 vintage, reference year 2020), where $A_{cs,c's'}$ represents purchases from sector s in country c by sector s' in country c' . Direct interruption $\beta^{(0)}$ propagates through the Leontief inverse:

$$\beta = (I - DA^T)^{-1} \beta^{(0)}$$

where D is a diagonal matrix of pass-through rates. The Leontief inverse captures both first-round and higher-order propagation effects. For data centers, the empirical multiplier is approximately $\times 11$: for every dollar of direct business interruption, eleven dollars of supply chain value are at risk. Critical dependence on geographically concentrated semiconductor fabrication (Taiwan 54%, South Korea 17%) and power equipment manufacturing (transformer lead times 12–24 months) amplifies this effect.

This formulation reveals that a facility's apparent geographic safety may be misleading. The Leontief analysis shows that 98–99% of BI propagates through supply chains rather than from direct on-site hazards.

Scenario Architecture

The analysis crosses four dimensions of uncertainty. **Physical pathways** (RCP 2.6, 4.5) determine temperature trajectories, extreme event frequency, and flood probabilities from downscaled CMIP projections at facility coordinates. **Transition policies** (BAU, Net Zero following NGFS) set carbon price trajectories and grid decarbonization pace. **AI growth scenarios** (LTG, SAI, AWB) determine the scale and geography of prospective infrastructure. **Adaptation strategies** (Baseline, Selective, Proactive) represent operator choices that modify realized ClimVaR.

The first two dimensions represent external uncertainty; the third determines the asset base; the fourth is the strategic variable. Three adaptation strategies are modeled. **Baseline**: current practice, no additional climate investment. **Selective**: targeted intervention on the highest-Shapley-value channel per facility (“one champion action”). **Proactive**: comprehensive investment across all five channels. Risk reductions across channels are multiplicative, not additive, because the channels interact through the Shapley structure.

Database

The **installed base** comprises 8,572 geocoded facilities totaling 90.8 GW across 120 countries. Sources include industry databases, regulatory filings, and satellite imagery validation. Facility capacities range from 15 kW to 300 MW (median 3.0 MW). Each facility carries 29 documented financial parameters: revenue, asset value, COGS ratio, OPEX ratio, tax rate, WACC, PUE, carbon intensity, cooling energy ratio, outdoor salary ratio, and insurance parameters.

Prospective capacity represents the AI-era pipeline through 2050, constructed from four source pools ordered by decreasing certainty: Epoch AI GPU cluster database (62 facilities, 6.3 GW, confirmed deployments), Tier 2 announced projects (60 facilities, 6.0 GW), and two probabilistic pools allocated via regional demand models calibrated against IEA projections.

Three nested scenarios select subsets: LTG draws 615 facilities (62.9 GW), SAI draws 1,195 (121.1 GW), AWB draws 2,939 (295.6 GW). Nestedness holds by construction: $LTG \subset SAI \subset AWB$. Under AWB, the total portfolio reaches 11,511 facilities (386.4 GW).

Asset valuation follows regional benchmarks: \$9M/MW for installed base, \$28–42M/MW for AI-tier facilities (reflecting higher power density and equipment cost). Revenue multipliers for AI workloads ($\times 2.0$) are calibrated against operator disclosures. Country-level parameters (tax rate, WACC, grid carbon intensity) draw from OECD, Damodaran, and IEA databases. All assumptions are stated explicitly in Appendix D. The going-concern multiplier is $\times 1.35$, accounting for terminal value beyond the explicit 25-year forecast horizon.

Exhibit 3. Facility coverage by region and AI growth scenario.

Region	Inst.	GW	+LTG	GW	+SAI	GW	+AWB	GW
United States	2,383	30.3	2,620	54.0	2,831	75.1	3,464	138.4
China	1,284	25.1	1,395	36.2	1,522	48.9	1,902	86.9
Europe	2,019	15.4	2,101	23.6	2,170	30.5	2,376	51.1
Asia-Pacific	1,593	11.6	1,692	22.9	1,804	34.3	2,145	68.5
Rest of World	1,293	8.5	1,379	17.1	1,440	23.2	1,624	41.6
Total	8,572	90.8	9,187	153.7	9,767	212.0	11,511	386.4

Installed base: industry databases, regulatory filings, satellite validation. Prospective: Epoch AI, Tier 2 announced, probabilistic allocation (IEA calibrated). LTG/SAI/AWB nested by construction.

This chapter presents four findings at macroeconomic scale and three facility case studies. All figures use the BAU climate-economic pathway and MEAN estimates unless noted otherwise.

Exhibit 4. Panoramic results: ClimVaR across asset classes, AI scenarios, and adaptation strategies (BAU/MEAN).

Portfolio	V ₀ (\$B)	Baseline	Selective	Proactive	Reduction
Installed base	1,012	388 (38%)	305 (30%)	238 (24%)	39%
United States	393	108 (28%)	89 (23%)	72 (18%)	33%
Europe	201	107 (53%)	81 (40%)	60 (30%)	44%
China	198	111 (56%)	86 (44%)	67 (34%)	40%
APMEA	220	62 (28%)	49 (22%)	39 (18%)	37%
Installed + LTG	1,992	1,008 (51%)	821 (41%)	669 (34%)	34%
Installed + SAI	2,849	1,669 (59%)	1,366 (48%)	1,134 (40%)	32%
Installed + AWB	5,590	3,671 (66%)	3,006 (54%)	2,536 (45%)	31%

Baseline = no adaptation. Selective = one champion action per facility. Proactive = comprehensive adaptation. ClimVaR% in parentheses = $\text{ClimVaR} / V_0$. Max reduction = $(\text{Baseline} - \text{Proactive}) / \text{Baseline}$. A ClimVaR of 38% means that risk-adjusted present value over 25 years is 38% below climate-naïve valuation.

Finding 1: Aggregate Exposure

Climate risk erodes 38% of current data center asset value _____

The installed base of 8,572 facilities—valued at \$1,012 billion—faces \$388 billion in cumulative climate-adjusted value erosion under the BAU pathway: a ClimVaR intensity of 38.3%. Five channels contribute: business interruption leads at 31%

(\$121B), followed by carbon costs 28% (\$107B), physical damages 22% (\$84B), cooling 19% (\$73B), and heat productivity 1% (\$2B).

The channel decomposition reveals a critical structural feature: supply chain BI (\$111B) exceeds direct BI (\$10B) by a factor of eleven—the Leontief propagation channel dominates the interruption risk. Tropical wind drives half of global supply chain BI (\$59B of \$111B), reflecting semiconductor and power equipment manufacturing concentration in typhoon-exposed East

Asia. Carbon costs are 99% Scope 2 (\$106B of \$107B), confirming that grid carbon intensity and PPA strategy are the primary levers for transition risk.

With prospective AI infrastructure, total ClimVaR scales to \$1.0 trillion (LTG), \$1.7 trillion (SAI), and \$3.7 trillion (AWB). ClimVaR *intensity* rises with AI scale—from 38% on the installed base to 51–66% with AI additions—indicating that prospective capacity carries higher risk intensity per GW than the legacy fleet.

Finding 2: Geographic Dispersion

Location and channel composition drive a twofold spread in risk

ClimVaR intensity ranges from 27.6% in the United States to 55.8% in China, with Europe at 53.1% and APMEA at 28.2%. The dominant risk channel shifts by region. The United States is BI-dominated (45% of regional risk), reflecting capacity concentration in Texas and Virginia where grid stress and extreme weather intersect. Europe is carbon-led (42%), driven by aggressive carbon pricing trajectories (€50–100/tonne today, projected €150–300 by 2040) applied to grids still partially dependent on fossil generation; the PPA calibration reduced Europe's carbon exposure by 47% from raw calculations. China shows dual exposure: physical damages (30%) and carbon costs (33%) contribute almost equally, combining flooding in

the Yangtze corridor, typhoons in southern provinces, and a coal-dependent grid facing uncertain transition. APMEA is damage-led (29%), reflecting coastal and seismic exposure across South and Southeast Asia.

Within these regional profiles, further disaggregation by hazard type reveals that river flood (\$49B) and coastal exposure (\$30B) together account for 94% of global physical damage—hydrological hazards, not wind or fire, drive the δ channel. APMEA's damage profile is dominated by sea-level exposure (\$13B of \$18B), while China faces a dual river-and-coastal threat.

Finding 3: AI Amplification and Structural Inversion

AI infrastructure inverts risk from carbon-dominated to physically-dominated

On the installed base, physical risk channels (BI + damages + cooling + heat) account for 72% of total ClimVaR, with carbon at 28%. When AI infrastructure is added, the composition shifts: physical channels account for 92–94% of ClimVaR on new AI-tier facilities across all three growth scenarios; carbon drops to 6–7%. This is not because carbon costs decrease—it is because physical risks amplify non-linearly with AI infrastructure addition.

The amplification is measurable. From the installed base (90.8 GW) to the AWB portfolio (386.4 GW), capacity grows $\times 4.3$, but physical risk grows $\times 11$ —a factor of $\times 2.6$ per GW added. Three mechanisms drive this: geographic concentra-

tion (AI facilities cluster in regions combining cheap power with elevated climate exposure), power density (30–100+ kW/rack generating more waste heat, making cooling failures more consequential), and operational criticality (non-interruptible AI training runs where interruption invalidates weeks of compute).

Under the AWB scenario, Europe reaches 93.5% ClimVaR intensity—near the destruction threshold where climate risk erodes virtually all asset value. Under Net Zero Transition pathways, carbon costs multiply $\times 6.2$ on the installed base (\$107B to \$661B). Only China reaches negative residual value ($V_{\text{resid}} = -\$93\text{B}$) under NZT with PPA-calibrated emissions.

Legacy assets face a carbon problem. AI-era assets face a physics problem. Net-zero commitments address 28% of installed base risk but less than 8% of AI-era risk. The adaptation strategies required for each are fundamentally different.

Finding 4: Adaptation Value

Proactive adaptation reduces risk by 30–39% with positive returns across all scenarios

On the installed base, Proactive adaptation reduces aggregate ClimVaR from \$388B to \$238B—a 39% reduction (\$150B protected). Europe shows the highest responsiveness (44% reduction), the US the lowest (33%), reflecting the carbon channel's high sensitivity to adaptation in European markets. The Selective strategy—one champion action per facility—captures 56% of Proactive's value at lower capital outlay.

Adaptation value scales with AI deployment. Under AWB, Proactive protects \$1,063B of value; Selective alone protects \$609B. Three adaptation pathways diverge within two years of implementation: Proactive reduces ClimVaR intensity from 41%

to 31% within that window, while Baseline remains above 45%. The front-loaded nature of adaptation value—driven by carbon channel response to PPA procurement—means that delayed action forfeits the most valuable early-year risk reduction.

Under NZT pathways, adaptation value increases further: NZT/Proactive achieves 53% reduction versus NZT/Baseline. Under NZT/Baseline, the installed base exceeds 100% ClimVaR intensity by 2035; under NZT/Proactive, it stays below 56%. Adaptation is valuable under every scenario, but its value *increases with climate policy ambition*.



Three Facility Case Studies

Translating aggregate patterns into site-level decisions

Three prospective AI data centers—each modeled at 100 MW to enable direct comparison—produce ClimVaR intensities ranging from 37.6% to 167%, a 4.5× dispersion. ClimVaR intensity is scale-independent; the channel composition and risk profile are identical at any capacity.

Exhibit 5. Three prospective AI facilities: site characteristics and ClimVaR decomposition (BAU/MEAN).

	Abilene, TX	Valence, FR	Guangdong, CN
Operator / Project	OpenAI Stargate	Sesterce sovereign	Pearl River Delta
Capacity	2,800 MW	100 MW	100 MW
PUE	1.15	1.12	1.28
β — BI	\$384M (43%)	\$374M (11%)	\$276M (53%)
γ — Cooling	\$336M (37%)	\$459M (13%)	\$96M (18%)
δ — Damage	\$143M (16%)	\$2,678M (75%)	\$29M (6%)
CC — Carbon	\$28M (3%)	\$42M (1%)	\$113M (22%)
ξ — Heat	\$11M (1%)	\$6M (<1%)	\$4M (1%)
Total ClimVaR	\$902M	\$3,560M	\$517M
Intensity	46.5%	167%	37.6%

V8 model results. Enterprise values calculated at facility level using regional financial parameters and 1.35× going-concern multiplier. BI includes both direct and supply chain (Leontief) components.

Abilene, Texas — The Power Fortress

Business interruption accounts for 43% of site ClimVaR (\$384M). The Leontief decomposition shows that virtually all BI (>99%) propagates through the supply chain: tropical wind (56%) and storms (24%) affecting upstream semiconductor and power equipment suppliers in East Asia dominate the interruption profile. ERCOT grid fragility contributes to direct operational risk but represents a smaller contribution than the indirect exposure. Cooling at 37% (\$336M) reflects West Texas ambient temperatures combined with AI compute heat density. Physical damage at 16% is dominated by wildfire (97% of δ). Carbon costs remain low (3%) but would escalate sharply under NZT: ×15 to \$427M, raising total ClimVaR to \$1.0B.

Valence, France — The River Flood Trap

The highest European ClimVaR intensity at 167%—risk exceeds enterprise value. Physical damage from river flooding accounts for 75% of total ClimVaR (\$2.7B). The Sesterce campus sits in the Rhône flood plain, and the damage function reflects both the probability and severity of fluvial hazards at these

coordinates. Cooling at 13% (\$459M) reflects rising temperatures in the Rhône Valley. The Shapley decomposition within β shows drought at 23%—above the global average—reflecting the valley’s chronic water stress affecting upstream suppliers. A ClimVaR exceeding 100% does not mean the facility is destroyed tomorrow; it means that the discounted lifetime losses exceed the current enterprise value.

Guangdong, China — The Triple Threat

A balanced risk profile at 37.6% intensity: BI (53%, \$276M), carbon (22%, \$113M), and cooling (18%, \$96M) combine in a triple exposure. BI is dominated by tropical cyclone risk (60% of β) propagating through supply chains (98% indirect). Carbon costs reflect Guangdong’s coal-heavy grid (carbon intensity 0.72 tCO₂/MWh) combined with rising carbon price trajectories. Physical damage is moderate (6%, \$29M)—dominated by wildfire (79%) rather than the flooding that characterizes the Pearl River Delta’s reputation. This site requires comprehensive Proactive strategies addressing all three dominant channels simultaneously.



At all three sites, 98–99% of business interruption propagates through supply chains rather than from direct on-site hazards. A facility’s apparent geographic safety may tell little about its actual interruption exposure. These two layers—hazard attribution and propagation pathway—are absent from traditional risk assessments and enable adaptation to be differentiated by site rather than applied uniformly.

The results in Chapter III establish the scale, geography, composition, and reducibility of climate risk for data center infrastructure. This chapter interprets those findings against existing literature, examines the mechanisms that drive risk amplification, interrogates key methodological choices, and draws implications for four stakeholder groups.

The \$388 Billion Baseline in Context

The US\$388 billion ClimVaR on the installed base (38.3% of enterprise value) can be situated against three reference points. The WEF/S&P aggregate of US\$3.3 trillion in cumulative costs by 2055 operates at a different analytical level—aggregate, not facility-level, and without channel decomposition—but the order of magnitude is consistent when discounting, scope, and horizon differences are accounted for. XDI's 2025 physical hazard assessment covers comparable facility counts (8,868 vs. 8,572) but reports structural damage metrics (MVAR%) rather than enterprise value loss; our framework extends this by adding cooling stress, carbon pricing, business interruption with supply chain propagation, and heat productivity—channels that collectively account for 65–75% of total ClimVaR depending on region. Dietz et al. (2016) estimated 1.8% of global financial assets at risk from climate change under a 2.5°C warming path; the 38% intensity we find for data centers reflects the sector's higher ex-

posure to the specific channels—energy costs, uptime requirements, supply chain concentration—through which climate affects this asset class.

The gap between 38% enterprise value erosion and zero in standard DCF valuations quantifies the mispricing. Current disclosures under TCFD and CSRD do not treat facility-level climate risk as explicit erosion of enterprise value. A portfolio of data center assets with a disclosed ClimVaR intensity of 38% should command a different price from one at 15%—yet no disclosure framework currently requires or enables this distinction.

The Supply Chain Multiplier

The $\times 11$ ratio of supply chain to direct business interruption is the most consequential finding for risk management practice. At all three case study sites, 98–99% of BI propagates through supply chains rather than from direct on-site hazards. A facility's apparent geographic safety may tell little about its actual interruption exposure.

This finding aligns with Verdantix's (2024) identification of supply chain risk quantification as the weakest capability across all 19 climate risk vendors benchmarked. It connects to documented concentration risks: Taiwan accounts for over 60% of advanced semiconductor fabrication; large power transformers carry 12–24 month lead times from a limited number of manufacturing sites; high-bandwidth memory production concentrates in South Korea. A single typhoon disrupting TSMC

operations would propagate through every data center in the global portfolio via the Leontief inverse. The January 2026 Virginia winter storm demonstrated a related mechanism at smaller scale: grid stress propagated through interconnected facilities faster than backup systems could respond, producing correlated losses across a geographic cluster.

The Leontief framework captures first-order and higher-order propagation, but assumes fixed technical coefficients—firms cannot reroute sourcing under stress. Long-duration disruptions are partially attenuated by adaptive behavior not captured in the model. The $\times 11$ multiplier should therefore be interpreted as an upper bound for prolonged disruptions and a reasonable estimate for acute shocks.

The Structural Inversion

The shift from carbon-dominated risk (72%/28% on the installed base) to physically-dominated risk (92–94%/6–8% on AI-era facilities) has direct implications for climate strategy. Net-zero commitments—which address the carbon component through PPAs, grid decarbonization, and on-site generation—are necessary for the installed base but insufficient for AI infrastructure. The climate policy dividend is asymmetric: under the Net Zero Transition pathway, installed base ClimVaR drops by 32%, but AI-era ClimVaR drops by only 12–18%, because the residual physical risk is irreducible through decarbonization alone.

Three mechanisms drive the amplification. Geographic concentration: AI facilities cluster in regions combining cheap power with fast permitting—Texas, Virginia, parts of China and South-east Asia—precisely the regions with elevated climate exposure. Power density: AI racks at 30–100+ kW/rack generate more waste heat per square meter, making cooling failures more con-

sequential. At a 5 kW/rack legacy facility, a cooling failure is an inconvenience; at a 60 kW/rack AI cluster, it triggers thermal shutdown within minutes. Operational criticality: AI training runs are non-interruptible workloads where a business interruption event does not merely pause revenue—it invalidates weeks of compute.

The $\times 2.6$ physical risk amplification per GW combines genuine physical amplification with a valuation effect: AI facilities carry higher asset value per MW (\$28–42M vs. \$9M for installed base), amplifying the financial impact of identical physical hazards. Separating these two components—physics from valuation—matters for adaptation: the physical component responds to engineering interventions (cooling, structural resilience), while the valuation component responds to financial structuring (insurance, diversification).

Adaptation: Investment Logic and Residual Risk

The 30–39% reduction achievable through Proactive adaptation, with positive NPV under every scenario, establishes adaptation as an investable proposition. But the residual 61% of ClimVaR that Proactive cannot eliminate deserves equal attention: it comprises extreme physical events (tail risk beyond facility-level mitigation), supply chain concentration (geographic dependencies that individual operators cannot diversify away), and thermodynamic limits (ambient temperature constraints on cooling that no technology fully overcomes in the hottest climates).

Two investment logics emerge from the trajectory analysis. The Selective strategy—adapting only new builds—captures 56% of Proactive's value at lower capital outlay and is justified under every scenario including the most benign. It targets the highest-leverage interventions (PPAs on new builds, liquid cooling at design stage, climate-aware siting) where marginal cost is low-

est. The Proactive strategy adds value through legacy retrofit, structured adaptation programs, and portfolio-level geographic rebalancing—measures that are more expensive per unit of risk reduced but address the 60–70% of installed capacity that will still operate legacy infrastructure through 2040.

The choice between strategies depends on the operator's position: for hyperscalers building primarily new capacity, Selective captures most of the value. For colocation providers and enterprise owners managing existing portfolios, Proactive addresses the installed base that Selective leaves untouched. Under NZT pathways, where carbon pricing accelerates to €150–300/tonne by 2040, the additional investment in legacy decarbonization generates the largest return. Adaptation is valuable under every scenario, but its value increases with climate policy ambition—operators who invest today are hedged against the transition scenario where that investment pays the most.

Implications by Stakeholder

For capital markets. Current valuations do not price climate risk. The 38% average intensity masks a twofold dispersion across regions and a 4.5× dispersion across individual facilities. Metrics that link location, risk channels, and scenarios to asset value loss—and that can be embedded in DCF models—would make this exposure visible. The framework's facility-level architecture produces outputs directly compatible with TCFD disclosure requirements.

For operators. The Shapley decomposition identifies which risk channel dominates at each facility, enabling targeted rather than uniform investment. The case studies demonstrate that identical capital deployed at Abilene (supply chain diversification + cooling), Valence (flood protection), and Guangdong (grid decarbonization + supply chain) generates different risk reductions—a dimension invisible without channel-level attribution. Data centers do not fail in isolation: they fail with grids and supply chains. The prospective capacity whose design has not been finalized represents the remaining degrees of freedom.

For insurers. The ×11 supply chain multiplier explains why facility-level loss ratios underestimate systemic exposure. Correlated losses across interconnected facilities—driven by shared grids, shared supply chains, and shared climate trajectories—require portfolio-level underwriting models that current practice does not provide. The Shapley + Leontief decomposition offers a structure for differentiating premiums by channel composition rather than aggregate hazard score.

For policymakers. Data centers have shifted from commercial real estate to strategic infrastructure, but their climate exposure remains weakly integrated into permitting, prudential oversight, and critical-infrastructure policy. Climate-adjusted siting rules, infrastructure stress tests, and public risk-sharing mechanisms are needed. The concentration of AI investment in climate-vulnerable locations—driven by power cost and permitting speed rather than resilience—represents a policy failure that the market alone is unlikely to correct, because the externalities (grid strain, water competition, systemic interruption risk) are not priced in individual facility economics.

Chapter V

Conclusions

Summary of Contributions

This paper has developed and applied a climate-adjusted valuation framework to 8,572 installed data centers (90.8 GW) and up to 2,939 prospective AI-era facilities across 120 countries. The framework integrates physical hazard projections, carbon pricing trajectories, and facility-level DCF analysis through five risk channels, with Shapley decomposition for hazard attribution and Leontief input-output analysis for supply chain propagation.

Four findings emerge. Aggregate exposure of US\$388B on the installed base (38.3% of enterprise value), scaling to US\$1.0–3.7T with AI infrastructure. A twofold geographic dispersion in risk intensity with distinct channel compositions by region. A $\times 2.6$ physical risk amplification for AI infrastructure per GW,

inverting the risk composition from carbon-dominated (72%) to physically-dominated (92–94%). And a 30–39% reduction achievable through proactive adaptation with positive net present value under every scenario tested. Three facility case studies confirm a $4.5\times$ dispersion at site level and demonstrate that effective adaptation must be differentiated by location.

The principal contribution is methodological: the combination of entity-level climate valuation, hazard attribution, and supply chain propagation applied to a sector where no comparable framework existed. The documented assumptions and modular architecture make the framework applicable to other infrastructure sectors.

Limitations

Climate projection uncertainty. Physical projections carry inherent uncertainty at facility-specific spatial scales. The scenario structure (RCP 2.6/4.5 $\times$ NZT/BAU) spans a wide range, but tail risks—tipping point cascades, correlated infrastructure failures, compound events exceeding historical analogs—may

exceed modeled bounds. The January 2026 Virginia winter storm illustrates the type of correlated event that the framework captures directionally but cannot predict with precision.

Archetype-based calibration. All facilities within a region and tier share financial parameters (PUE, PPA factor, WACC, cost

ratios), calibrated from JLL, Synergy Research, CBRE, and IEA benchmarks. This enables coverage of the full database but does not capture facility-specific characteristics: construction materials, elevation, cooling system vintage, backup generator capacity. The framework is most reliable for portfolio-level and regional comparison; individual facility estimates carry uncertainty proportional to their divergence from the archetype.

Prospective infrastructure. The 615–2,939 prospective facilities are based on announced, planned, and speculative projects. Actual buildout will differ in timing, scale, and location. ClimVaR estimates for prospective infrastructure should be read as indicative of the risk profile under each scenario, not as point predictions.

Future Research

Empirical calibration. The January 2026 Virginia winter storm provides a natural experiment for calibrating the β channel against observed losses. Systematic collection of post-event operational and financial data—downtime duration, revenue impact, insurance claims, recovery costs—would enable model validation. Collaboration with operators and insurers would be required.

Asset-specific resolution. Replacing regional archetypes with building-specific data (construction materials, floor elevation, cooling system type, generator capacity) would improve individual facility estimates. Integration of satellite imagery, planning databases, and operator-provided attributes could achieve this without sacrificing database scale.

Water-energy-climate nexus. Liquid cooling reduces energy consumption but increases water consumption. In water-stressed regions—the US Southwest, parts of India, northern China—this substitution may exchange one climate vulnerability for another. Integrating water availability projections with cooling technology trajectories would close a gap in the current analysis.

Real options for adaptation timing. The current framework treats adaptation as a binary decision. A real-option extension would value the flexibility to delay while preserving the ability

Adaptation modeling. The Selective and Proactive strategies apply uniform reduction rates by channel. In practice, adaptation effectiveness varies by facility age, operator capability, and local conditions. The $\times 2.6$ amplification combines genuine physical amplification with a valuation effect (AI facilities carry higher asset value per MW), as discussed in Chapter IV.

Scope of risk. The framework quantifies private financial risk. It does not capture systemic externalities: grid strain from concentrated AI load, water resource competition, community displacement, or cascading economic effects of simultaneous service interruptions. The US\$388 billion figure represents a lower bound on total economic exposure.

to act—relevant for legacy facilities where the 15–20% versus 40–60% CAPEX differential between greenfield and retrofit suggests significant option value in early commitment.

Expected Shortfall and tail risk. Extending ClimVaR to Expected Shortfall (CVaR)—the conditional expectation of losses beyond a given quantile—would provide a risk metric better suited to insurance pricing and prudential regulation, where tail losses dominate decision-making.

Cross-sector generalization. The combination of entity-level ClimVaR, Shapley attribution, and Leontief propagation provides a transferable methodology. Manufacturing, energy, and transportation infrastructure face analogous challenges. Extension to multi-sector portfolios—particularly at points of interdependency (data centers $\times$ power grids $\times$ telecommunications)—would capture systemic risks that sector-specific analysis underestimates.

Dynamic scenario integration. Making scenario pathways dynamic—time-varying carbon prices responding to policy signals, adaptive operator strategies triggered by observed losses, evolving MRIOT coefficients reflecting supply chain restructuring—would shift the framework from comparative statics to genuinely dynamic simulation of infrastructure investment under evolving climate conditions.

Closing

Under the Abundance Without Boundaries scenario, over US\$4,600 billion in prospective AI infrastructure will be built within the next decade, operate through mid-century, and face climate conditions that diverge progressively from the historical baselines that informed their design. The computational systems housed in these facilities model climate futures, optimize

energy systems, and analyze risk across every sector of the global economy. Applying climate risk quantification to the sector's own infrastructure addresses a documented analytical gap.

Every data center built today embeds assumptions about climate futures. The framework developed here makes those

assumptions explicit and their financial consequences quantifiable. The prospective infrastructure whose design has not been finalized—between 63 and 296 GW depending on growth

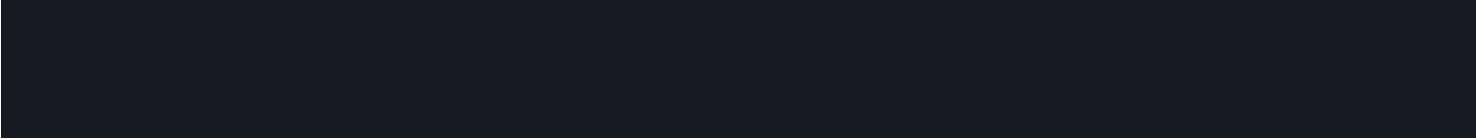
scenario—represents the remaining degrees of freedom. The analytical basis for informing those decisions now exists.

**Climate risk for data centers is unpriced, not unrecognized.
This paper provides the pricing framework.**

References

- [1] Allianz Commercial (2025). *The Data Center Construction Boom: Risks and Insurance Implications*. Allianz Commercial.
- [2] Associated Press (2025). Memories of deadly winter blackouts haunt the Texas data center boom. *The Spokesman-Review*.
- [3] Botzen, W.J.W., et al. (2019). Economic impacts of natural disasters. *Review of Environmental Economics and Policy*, 13, 167–188.
- [4] Brealey, R.A., Myers, S.C., Allen, F. (2020). *Principles of Corporate Finance*. McGraw-Hill.
- [5] Connell, J. (2026). *PJM's Updated 20-Year Forecast Continues To See Significant Long-Term Load Growth*. PJM Inside Lines.
- [6] Damodaran, A. (2012). *Investment Valuation*. Wiley.
- [7] Data Center Dynamics (2012). *Hurricane Sandy: How Data Centers Weathered the Storm*.
- [8] Data Center Dynamics (2025). Over 6% of data centers face high physical climate risk, report finds. *DCD*.
- [9] Data Center Knowledge (2021). Most Texas data centers weathered the storm. It didn't go smoothly. *DCK*.
- [10] Dawson, R., et al. (2018). A systems framework for national assessment of climate risks to infrastructure. *Phil. Trans. R. Soc. A*, 376, 20170298.
- [11] Deloitte (2025). *Can US Infrastructure Keep Up with the AI Economy?*. Deloitte Center for Energy and Industrials.
- [12] Dietz, S., et al. (2016). Climate value at risk of global financial assets. *Nature Climate Change*, 6, 676–679.
- [13] Dietz, S., et al. (2021). Economic impacts of tipping points in the climate system. *PNAS*, 118, e2103081118.
- [14] Donovan, P. (2025). *AI Workload Deployment in Data Centers: Retrofit, Outsource or Build New?*. Schneider Electric.
- [15] ECB, EIOPA (2024). *Policy Options to Reduce the Climate Insurance Protection Gap*. ECB/EIOPA.
- [16] Epelbaum, T. (2025). ClimVaR: Discounted cash flow approach on climate risks. Schneider Electric Advisory Services.
- [17] Equinix (2025). *Global Tech Trends Survey 2025*. Equinix.
- [18] Eyring, V., et al. (2016). Overview of CMIP6. *Geoscientific Model Development*, 9, 1937–1958.
- [19] Forzieri, G., et al. (2018). Escalating impacts of climate extremes on critical infrastructures in Europe. *Global Environmental Change*, 48, 97–107.
- [20] Gallagher Re (2026). *Natural Catastrophe and Climate Report*. January 2026.
- [21] Gault, M., Williams, J. (2025). We mapped the world's hottest data centers. *Rest of World*.
- [22] Hexatronic (2025). *Hurricane Season 2024: Impact on Data Center and Fiber Infrastructure*.
- [23] Hogan Lovells (2025). France reforms natural catastrophe insurance regime. *Hogan Lovells Insurance Hub*.
- [24] Howland, E. (2026). Data centers were 40% of PJM capacity costs in last auction. *Utility Dive*.
- [25] IEA (2024). *Electricity 2024*. International Energy Agency.
- [26] IEA (2025). *Energy and AI*. International Energy Agency.
- [27] IEEFA (2025). *Projected Data Center Growth Spurs PJM Capacity Prices by Factor of 10*. IEEFA.
- [28] In, S.Y., Weyant, J., Manav, B. (2020). Pricing climate-related risks of energy investments. *SSRN*.
- [29] In, S.Y., et al. (2021). Pricing climate risks of energy investments: A comparative case study. *Energy Economics*.
- [30] In, S.Y., et al. (2022). Climate-related financial risk assessment on energy infrastructure investments. *SSRN*.
- [31] JLL (2024). *Global Data Center Outlook 2024*.
- [32] Karagiannakis, G., Panteli, M., Argyroudis, S. (2024). Fragility modeling of power grid infrastructure for climate change. *WIREs Climate Change*, 15, e866.
- [33] Kerkhofs, R., et al. (2025). An asset-level analysis of financial tail risks under extreme weather events. *Environmental Research: Climate*, 4, 015001.
- [34] Kjellstrom, T., et al. (2016). Heat and occupational health. *Annual Review of Public Health*, 37, 97–112.
- [35] Klöckl, B., Papaefthymiou, G. (2010). Time series models for power systems. *Electric Power Systems Research*, 80, 265–276.
- [36] Koks, E., et al. (2022). Impacts of coastal flooding and sea level rise on critical infrastructure. *Sustainable and Resilient Infrastructure*, 8, 237–255.
- [37] Leontief, W.W. (1986). *Input-Output Economics*. Oxford University Press.
- [38] Luo, T., et al. (2023). A framework to assess multi-hazard physical climate risk for power generation. *Communications Earth & Environment*, 4, 117.
- [39] Masanet, E., et al. (2020). Recalibrating global data center energy-use estimates. *Science*, 367, 984–986.
- [40] Miller, R.E., Blair, P.D. (2009). *Input-Output Analysis: Foundations and Extensions*. Cambridge University Press.
- [41] NERC (2025). *2025–2026 Winter Reliability Assessment*. North American Electric Reliability Corporation.

- [42] NGFS (2023). *NGFS Scenarios for Central Banks*. Network for Greening the Financial System.
- [43] NRDC, Citizens Utility Board (2025). *Rising Demand from Data Centers Driving Reliability, Cost Concerns*. NRDC.
- [44] Oughton, E., et al. (2023). Global vulnerability assessment of mobile telecommunications infrastructure to climate hazards. *Scientific Data*.
- [45] Owen, A.B., Prieur, C. (2017). On Shapley value for measuring importance of dependent inputs. *SIAM/ASA J. Uncertainty Quantification*, 5, 986–1002.
- [46] Paccou, R., Wijnhoven, F. (2024). *Artificial Intelligence and Electricity: A System Dynamics Approach*. Schneider Electric SRI.
- [47] Pasha, M.D.S., et al. (2025). Flood damages in an infrastructure climate risk stress test. *J. Infrastructure Policy and Management*.
- [48] Pasupuleti, M.K. (2022). Climate and flood risk in data centre strategy. *Int. J. Academic and Industrial Research Innovations*.
- [49] Pasupuleti, M.K. (2024). Climate and flood-aware data centre siting: A GIS methodology. *Int. J. Academic and Industrial Research Innovations*.
- [50] Pasupuleti, M.K. (2025). GIS-driven AI for climate-resilient data centre siting and operation. *Int. J. Academic and Industrial Research Innovations*.
- [51] Prime, T., et al. (2018). Protecting energy infrastructure against future climate change: A real options approach. *J. Ocean and Coastal Economics*, 5, Article 1.
- [52] Reuters (2026). Power prices surge as winter storm spikes demand in U.S. data center alley. *CNBC*.
- [53] Roncalli, T. (2024). Carbon pricing and portfolio management. *SSRN*.
- [54] S&P Global (2025). *Geopolitics of Data Centers: An AI Showdown That Will Reshape the World*. S&P Global Market Intelligence.
- [55] Shapley, L.S. (1953). A value for n -person games. *Contributions to the Theory of Games*, 2, 307–317.
- [56] Shehabi, A., et al. (2016). *United States Data Center Energy Usage Report*. Lawrence Berkeley National Laboratory.
- [57] Shorrocks, A.F. (2013). Decomposition procedures for distributional analysis: A unified framework based on the Shapley value. *J. Economic Inequality*, 11, 99–126.
- [58] Stapczynski, S. (2026). US calls on data centers to prepare backup power ahead of storm. *Bloomberg*.
- [59] Swiss Re Institute (2025). *Sigma 1/2025: Natural Catastrophes*. Swiss Re.
- [60] Synergy Research Group (2025). *Global Data Center Market Update Q1 2025*.
- [61] TechTarget (2024). *Uptime Institute 2024 Annual Outage Analysis: Weather-Related Failures on the Rise*.
- [62] TRG Datacenters (2024). *Lessons from Hurricane Harvey: Data Center Resilience in Houston*.
- [63] Trigg, M.A., et al. (2016). Global fluvial flood risk analysis. *Environmental Research Letters*, 11.
- [64] Tyron, F. (2025). *Data Centers: Critical Infrastructure in a Volatile World*.
- [65] Verdantix (2024). *Smart Innovators: Physical Climate Risk Solutions*. December 2024.
- [66] WEF, S&P Global (2025). *Data Centers and the Climate Challenge: A US\$3.3 Trillion Question*. World Economic Forum.
- [67] WWF (2025). *Tackling the Insurance Protection Gap: Leveraging Climate and Nature to Increase Resilience*. World Wildlife Fund.
- [68] XDI (2025). *2025 Global Data Centre Physical Climate Risk and Adaptation Report*. Cross Dependency Initiative.





Appendix A

Supply Chain Propagation

After Wassily Leontief (1906–1999)

The data for a particular year, set out in the form of a rectangular input-output table, describe the flows of goods and services among all sectors of an economy.

Business interruption at a data center does not stay at the facility. It propagates through supply chains—equipment suppliers, service providers, grid operators, and downstream customers. The Leontief input-output framework traces these linkages.

For data centers, the empirical multiplier is $\times 11$: for every dollar of direct business interruption, eleven dollars of supply chain value are at risk. This appendix derives the multiplier from sector-level input-output tables, validates it against insurance loss data, and discusses its sensitivity to geographic concen-

tration in semiconductor and high-bandwidth memory supply chains.

Contents

- A.1 Modified Leontief framework for digital infrastructure
- A.2 Sector-level calibration and data sources
- A.3 Multiplier derivation and validation
- A.4 Sensitivity to supply chain concentration



Appendix B

Risk Channel Decomposition

After Lloyd Shapley (1923–2016)

Each player receives a payoff equal to his average marginal contribution over all possible orderings of the players.

Climate risk for data centers is not a single hazard—it is a portfolio of five interacting channels. Because these channels interact non-additively (cooling stress amplifies business interruption; carbon pricing compounds with physical damage), simple summation double-counts risk.

The Shapley value from cooperative game theory provides a principled decomposition. For $n = 5$ risk channels, it evaluates all $2^5 = 32$ coalition combinations per facility, attributing each channel's marginal contribution to total ClimVaR. The decompo-

sition satisfies efficiency, symmetry, and the null player property.

Contents

- B.1 Shapley value formulation for risk attribution
- B.2 Computational implementation (32 coalitions)
- B.3 Interpretation: which channels drive exposure
- B.4 Adaptation targeting via marginal Shapley values



Appendix C

Adaptation as Portfolio Optimization

After Harry Markowitz (1927–2023)

Diversification is both observed and sensible; a rule of behavior which does not imply the superiority of diversification must be rejected both as a hypothesis and as a maxim.

Adaptation investment across a portfolio of data centers mirrors the logic of financial portfolio theory. Operators allocate finite capital across facilities with heterogeneous risk profiles, seeking the combination that maximizes risk reduction per dollar invested.

The framework models three strategies—Baseline, Selective, and Proactive—and demonstrates that risk reductions are multiplicative, not additive, because channels interact through the Shapley structure. The “one champion action” approach (Sele-

tive) targets the highest-Shapley-value channel per facility.

Contents

- C.1 Adaptation strategies: Baseline, Selective, Proactive
- C.2 Non-additivity of risk reductions
- C.3 NPV of adaptation investment by scenario
- C.4 Geographic allocation of adaptation capital



Appendix D

Data & Scenario Hypotheses

After Gro Harlem Brundtland (b. 1939)

Sustainable development is development that meets the needs of the present without compromising the ability of future generations to meet their own needs.

The empirical foundation rests on a geocoded database of 8,572 installed data centers across 120 countries (90.8 GW), linked to facility-level financial parameters, climate projections, and three AI-era growth scenarios. This appendix documents all data sources, calibration choices, and scenario assumptions—enabling reproducibility and sensitivity testing.

Three growth scenarios bracket the uncertainty of AI infrastructure deployment: LTG (conservative), SAI (mid-range), and AWB (upper bound). Two climate-economic pathways (BAU at RCP 4.5, Net Zero at RCP 2.6) span the range of policy ambi-

tion.

Contents

- D.1 Database construction: sources and validation
- D.2 Financial parameters by facility type
- D.3 Climate projection methodology (2026–2050)
- D.4 AI growth scenarios: LTG, SAI, AWB
- D.5 Going-concern multiplier calibration ($\times 1.35$)

